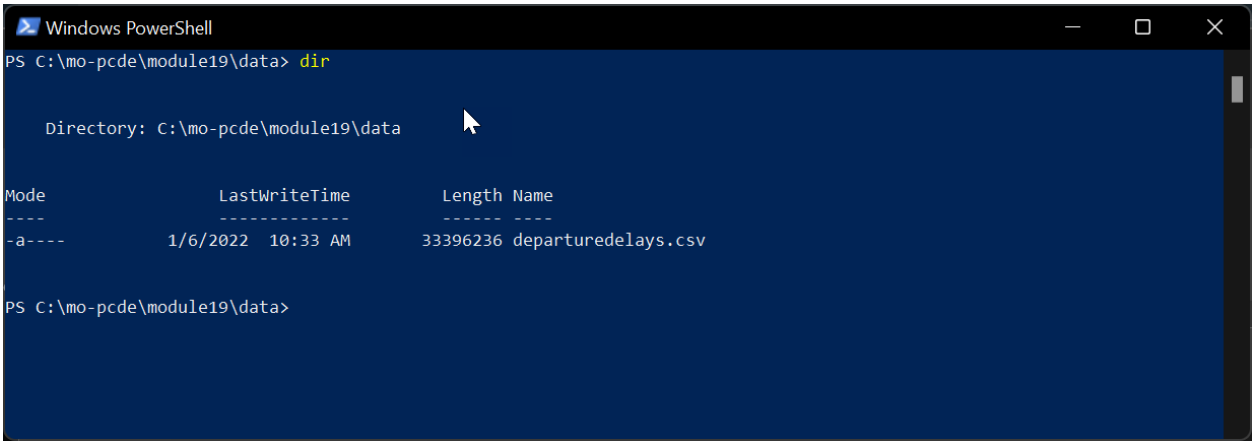
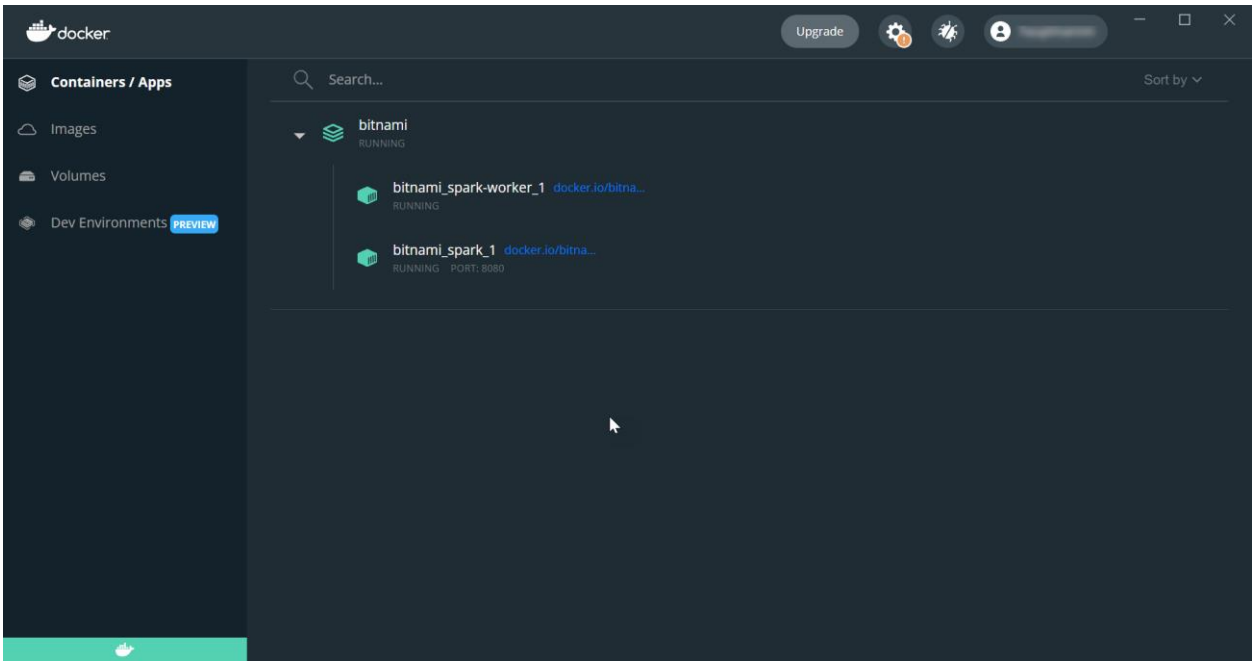
Activity 19.2 Solution File

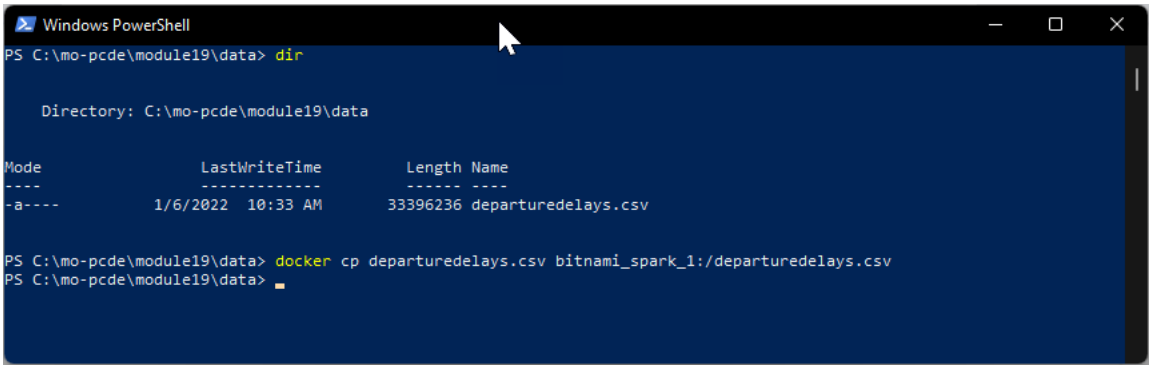
1.



2.

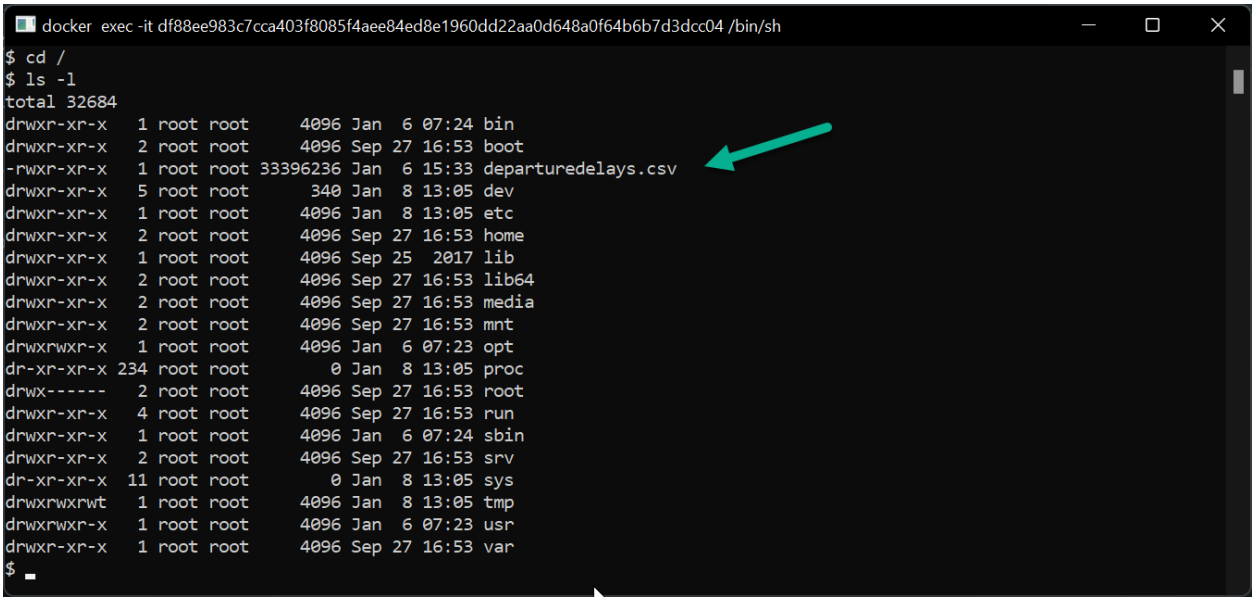


3.

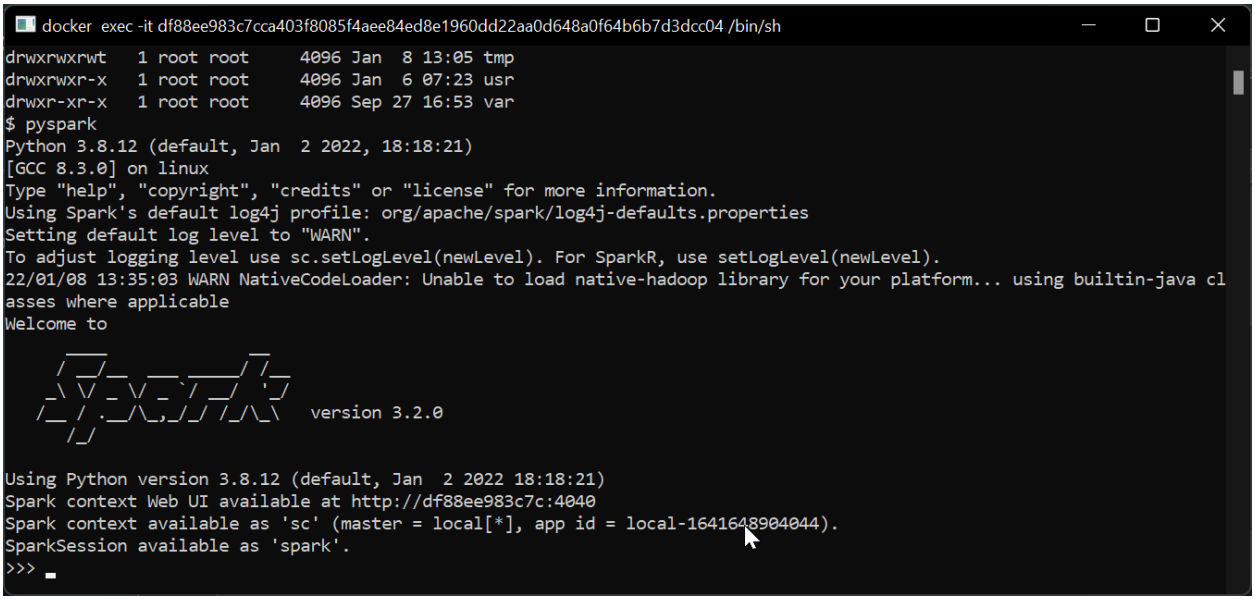


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4.

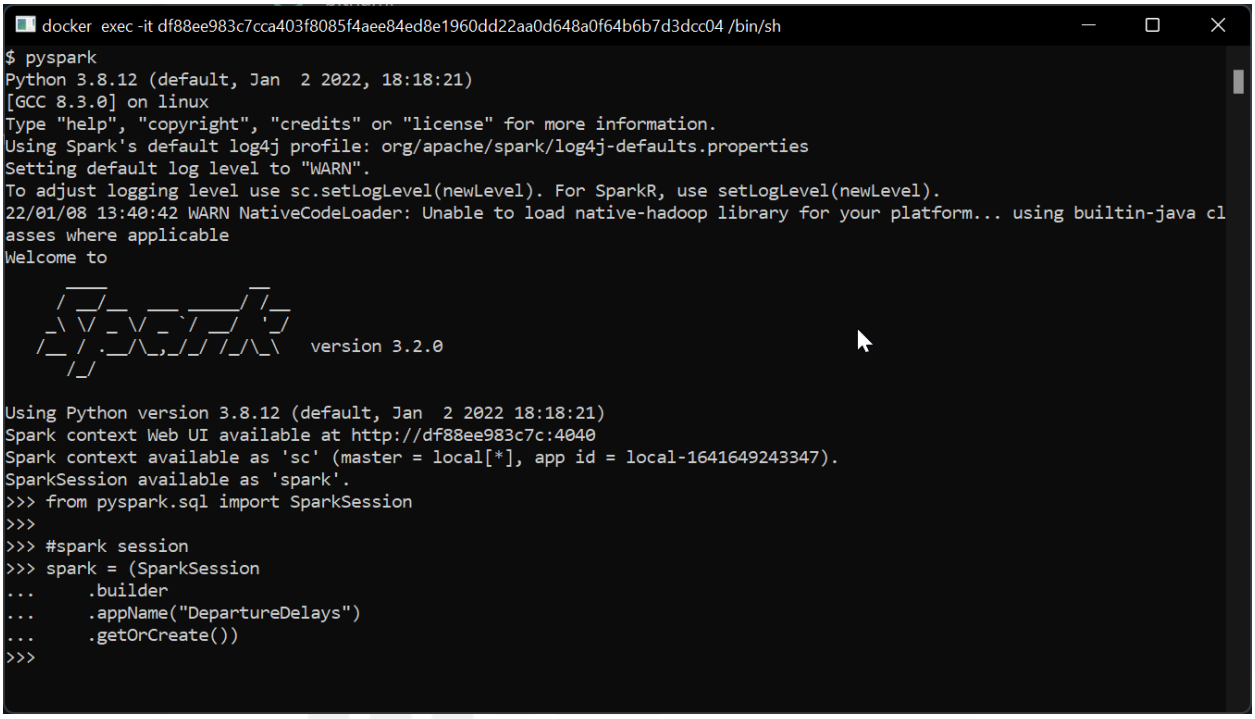


5.



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6.



7.



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8.

```
docker exec -it df88ee983c7cca403f8085f4aee84ed8e1960dd22aa0d648a0f64b6b7d3dcc04 /bin/sh
>>> spark.sql("""SELECT distance, origin, destination
... FROM delays_table WHERE distance > 1000
... ORDER BY distance DESC""").show(10)
+--------+------+-----------+
|distance|origin|destination|
+--------+------+-----------+
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
|    4330|   HNL|        JFK|
+--------+------+-----------+
only showing top 10 rows

>>>
```

9.

```
docker exec -it df88ee983c7cca403f8085f4aee84ed8e1960dd22aa0d648a0f64b6b7d3dcc04 /bin/sh
>>> spark.sql("""SELECT distance, origin, destination
... FROM delays_table WHERE delay > 120 and ORIGIN='SFO' and DESTINATION = 'ORD'
... ORDER BY delay DESC""").show(10)
+--------+------+-----------+
|distance|origin|destination|
+--------+------+-----------+
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
|    1604|   SFO|        ORD|
+--------+------+-----------+
only showing top 10 rows

>>>
```

10.

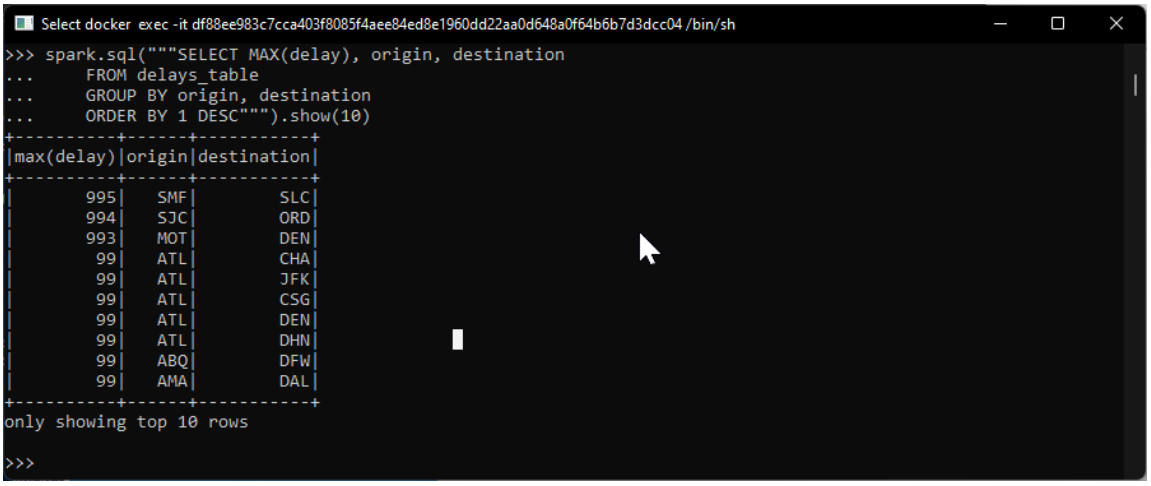
```
docker exec -it df88ee983c7cca403f8085f4aee84ed8e1960dd22aa0d648a0f64b6b7d3dcc04 /bin/sh

>>> spark.sql("""SELECT delay, origin, destination,
...     CASE
...             WHEN delay > 360 THEN 'Very long Delays'
...             WHEN delay > 120 AND delay < 360 THEN 'Long Delays'
...             WHEN delay > 60 AND delay < 120 THEN 'Short Delays'
...             WHEN delay > 0 AND delay < 60 THEN 'Tolerable Delays'
...             WHEN delay = 0 THEN 'No Delays'
...             ELSE 'Early'
...     END AS delay_length
...     FROM delays_table
...     ORDER BY origin, delay DESC""").show(10)
+-----+------+-----------+---------------+
|delay|origin|destination|   delay_length|
+-----+------+-----------+---------------+
|   92|   ABE|        ORD|    Short Delays|
|   91|   ABE|        DTW|    Short Delays|
|    9|   ABE|        ATL|Tolerable Delays|
|    9|   ABE|        ATL|Tolerable Delays|
|    9|   ABE|        ORD|Tolerable Delays|
|    9|   ABE|        ATL|Tolerable Delays|
|    9|   ABE|        ATL|Tolerable Delays|
|    9|   ABE|        ATL|Tolerable Delays|
|   89|   ABE|        DTW|    Short Delays|
|   88|   ABE|        ATL|    Short Delays|
+-----+------+-----------+---------------+
only showing top 10 rows

>>>
```

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11.



12.



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